

Answer the questions in the spaces provided on the question sheets.

Please give clear and rigorous answers.

Be to the point. Show your work.

Name: _____

ERP: _____

Question:	MCQs	Short Questions	OOP	Arrays	Recursion	Total
Marks:	20	15	10	10	10	65
Score:						

Question 1: MCQs 20 marks

For each of the following questions, circle the option that BEST answers the question (more than one option may be correct).

- (a) Consider the method signature **public static String foo(int i)**. Which of the following could be a correct method call? (Note: in all of the examples below **i** is of type **int**, and **s** is of type **String**).

A. **String foo(int i)**

C. **s = foo(i)**

B. **String foo(i)**

D. **foo(String s)**

- (b) For the loop given below, what is the final value of **i** when the loop is finished?

```
int i = 1;
int arr = {2, 3, 5};
while (i < arr.length) {
    i = i + 1;
}
```

A. 1

D. 3

B. 2

E. 4

C. You cannot tell just from the code segment provided

F. The loop will never exit – it is an infinite loop

(c) Which expression adds 1 to the element of array **arrayName** at index **i**, assuming the array is of type **int**?

- A. **++arrayName[i]**
- B. **arrayName++[i]**
- C. **arrayName[i++]**
- D. None of the above.

(d) The purpose of the following code segment

```
b = a + b;  
a = b - a;  
b = b - a;
```

where **a**, **b** are two **ints** is to

- A. Transfer the contents of **a** to **b**
- B. Transfer the contents of **b** to **a**
- C. Exchange the contents of **a** and **b**
- D. None of the above

(e) To enable *exception handling* on a block of code, one

- A. encloses it in a **try** block
- B. encloses it in a **throwable** block
- C. labels it, then inserts that label into the Java exception handler list
- D. must place the code into its own method and mark the method **throwable**
- E. enables a **trap** for that code

(f) Which of these statements about constructors is false?

- A. A constructor has no return type
- B. Its name must be the same as the class in which it is defined
- C. There must be exactly one constructor defined for a class
- D. Constructors are almost always declared as public
- E. They can appear anywhere in the class where it is legal to declare a method

(g) Which of the following statements about the **break** statement is false?

- A. The **break** statement is used to exit a repetition statement early and continue execution after the loop.
- B. A **break** statement can only break out of an immediately enclosing **while**, **do..while**, **for**, or **switch** statement.
- C. The **break** statement, when executed in a **while**, **for** or **do..while**, skips the remaining statements in the loop body and proceeds with the next iteration of the loop.
- D. Common uses of the **break** statement are to escape early from a loop or to skip the remainder of a **switch**.

(h) Consider the following two code segments:

```
int i = -1;
while (i < 20) {
    i++;
    System.out.println(i);
}
```

```
for (int i=0; i <= 20; i++) {
    System.out.println(i);
}
```

Which of the following statements is true?

- A. The output from these segments is not the same.
- B. The scope of the control variable `i` is different for the two segments.
- C. Both (A) and (B) are true.
- D. Neither (A) nor (B) is true.

(i) What is the output from the following Java program fragment? Read it carefully!

```
String S = ""; String T = "";
int i = 4;
for (i = 1; i <= 3; i++){
    S = S + "!";
}
for (i = 1; i < 4; i++){
    T = T + "*";
}
System.out.print(S);
System.out.println(T);
```

- A. (That is, the empty string, printed twice)
- B. *
- C. !***
- D. !****
- E. !!!***

(j) Which of the following expressions is equivalent to the boolean expression

`!(x < 5 && y != z)`

- A. `x > 5 || y != z`
- B. `x >= 5 && y == z`
- C. `!(x < 5) || (y != z)`
- D. `x >= 5 || y == z`
- E. `x < 5 && y == z`

Question 2: Short Questions 15 marks

(a) [10 marks] For each question below, circle the correct answer:

- i. TRUE FALSE A data type is a set of values and a set of operations on those values.
- ii. TRUE FALSE A class can define more than one instance method, but each instance method must have a different name.
- iii. TRUE FALSE An instance method can refer to the private instance variables of the invoking object, but cannot refer to the private instance variables of any other object.
- iv. TRUE FALSE A class can contain either static methods or instance methods, but not both.
- v. TRUE FALSE It is a compile-time error to define a local variable with the same name as an instance variable.
- vi. TRUE FALSE One array cannot contain both **int** and **double** elements.
- vii. TRUE FALSE **(int)(6 * Math.random())** will return a random integer between 1 and 6.
- viii. TRUE FALSE **java printOutput | out.txt** will redirect the output of **printOutput** to the file **out.txt**
- ix. TRUE FALSE Any code that can be written with multiple **if-else** statements could also be written with a **switch-case** statement.
- x. TRUE FALSE Any code that can be written with a **switch-case** statement could also be written with multiple **if-else** statements.

- (b) [5 marks] For each of the two functions below, give an intuitive description **in 20 words or less** of the function's purpose. In addition, **in 30 words or less** state whether the function will compile and run properly in all cases and what the error(s) will be otherwise.

i.

```
public static void one(int x) {  
    while (x != 0) {  
        System.out.println(x);  
        x--;  
    }  
}
```

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ii.

```
public static int two(char x) {  
    String keyboard = "abcdefghijklmnopqrstuvwxyz";  
    for (int i = 0; i < keyboard.length(); i++)  
        if (keyboard.charAt(i) == x)  
            return i;  
}
```

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Question 3: OOP 10 marks

- (a) [4 marks] Match the following Java keywords on the left to the most appropriate description from the right. Write one letter in each blank, referring to the descriptions below. Each letter will be used exactly once.

- | | |
|-------------------------|---|
| • class: _____ | A: defines something as being part of the API |
| • public: _____ | B: creates an instance |
| • static: _____ | C: hidden from clients in other <code>.java</code> files |
| • void: _____ | D: method name that is called when java is run from the command line |
| • main: _____ | E: return type of a method that does not return any value |
| • private: _____ | F: belongs to a class rather than to instances of that class |
| • this: _____ | G: contains definitions of methods and fields |
| • new: _____ | H: refers to the instance upon which the current method or constructor acts |

- (b) [2 marks] What is wrong with the constructor method in the following code?

```
public class Charge {
    private double rx, ry; // position
    private double q;      // charge

    public Charge(double x0, double y0, double q0) {
        double rx = x0;
        double ry = y0;
        double q = q0;
    }
    ...
}
```

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- (c) [4 marks] Write a class **Bug** that models a bug moving along a horizontal line. The bug moves either to the right or left. Initially, the bug moves to the right, but it can turn to change its direction. In each move, its position changes by one unit in the current direction.

Provide a constructor

```
public Bug(int initialPosition)
```

and methods

```
public void turn()
```

```
public void move()
```

```
public int getPosition()
```

Sample usage:

```
Bug bugsy = new Bug(10);
bugsy.move(); // Now the position is 11
bugsy.turn();
bugsy.move(); // Now the position is 10
```

[illegible]

Question 4: Arrays 10 marks

- (a) [2 marks] What elements does the array `numbers` contain after the following code is executed?
(Write the elements in the format: `{0, 1, 2, ...}`)

```
int[] numbers = new int[8];
numbers[1] = 3;
numbers[4] = 99;
numbers[7] = 2;

int x = numbers[1];
numbers[x] = 44;
numbers[numbers[7]] = 11; // uses numbers[7] as index
```

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.....

- (b) [2 marks] What is the output of the following code fragment:

```
int[][] table = new int[4][3];

for (int t = 0; t < 4; t++) {
    for (int i = 0; i < 3; i++) {
        table[t][i] = (t * 4) + i + 1;
        System.out.print(table[t][i] + " ");
    }
    System.out.println();
}
```

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.....
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- 11 11 7 9 16 4 1

would be considered identical. You will probably need one or more helper methods.

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Question 5: Recursion 10 marks

- (a) [4 marks] A palindrome is a word that reads the same forward and backward. The following program prints a special kind of palindrome.

```
public class Palindrome {  
    public static void main(String[] args) {  
        int N = Integer.parseInt(args[0]);  
        System.out.println(palindrome(N));  
    }  
    public static String palindrome(int i) {  
        if (i == 0) return "S";  
        if (i == 1) return "T";  
        return palindrome(i-2) + palindrome(i-1) + palindrome(i-2);  
    }  
}
```

- i. Give the string printed when this program is run with

% java Palindrome 3

.....

- ii. Give the string printed when this program is run with

% java Palindrome 4

.....

- iii. Give the length of the string printed for

% java Palindrome 7

.....

(b) [6 marks] Use recursion to implement a method

```
public static boolean find(String text, String str)
```

that tests whether a given `text` contains a string. For example, `find("Mississippi", "sip")` returns `true`.

Hint: If the `text` starts with the string you want to match, then you are done. If not, consider the `text` that you obtain by removing the first character.

This image shows a full page of dot grid paper. It features approximately 28 horizontal rows of small, evenly spaced dots. The dots are arranged in straight lines across the width of the page, providing a guide for writing or drawing without solid lines. The background is white, and the dots are a light gray color.